Deepan K S

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Summary

AI Engineer specializing in production Computer Vision, Agentic AI, and GenAI systems — real-time multi-stream pipelines (NVIDIA DeepStream, InsightFace, FAISS), LangGraph multi-agent RAG with LLM guardrails, and GPU-optimized inference.

Experience

Bhatiyani Astute Intelligence Pvt. Ltd Bangalore *Nov 2025 – Jun 2026*

Artificial Intelligence Intern

- Built real-time multi-camera attendance pipeline on NVIDIA **DeepStream** (nvinfer + nvtracker) with **InsightFace ArcFace** + dual-index **FAISS** recognition microservice via FastAPI and MongoDB; published events via RabbitMQ.
- Implemented entry/exit line-crossing counters, staff-customer proximity/dwell detection, unattended-customer alerts, and automatic pipeline recovery for 24/7 reliability.

Resilience Business Grids (RBG.AI) Coimbatore *Jul 2025 – Nov 2025*

Artificial Intelligence Intern

- Developed multi-modal security pipeline with custom **YOLOv8x** (11 bag classes), ViT-based gender classification, and anomaly detection (Abandoned Bag, Isolated People, Help Gesture) via MediaPipe + Deep SORT with Optical Flow camera stabilization.

Projects

Production Agentic RAG System — LLM Gateway, Guardrails & Multi-Agent Orchestration Live Demo |

- Deployed full-stack multi-agent RAG on AWS EC2: Streamlit frontend, Flask REST API, and **LangGraph** orchestration with intent router (CHITCHAT / RESEARCH / SAVE_HISTORY) and persistent MemorySaver across sessions.
- Built **SmartLLMGateway**: complexity-based routing to Groq Llama-3.1-8b / Llama-3.3-70b, automatic Gemini fallback, MD5 prompt caching (100% token savings on hits), and load-balanced API key rotation with per-query cost tracking.
- Implemented dual-layer **guardrail**: regex + LLM cognitive layer (prompt injection detection, fail-secure) and output hallucination check cross-referencing responses against a Pydantic FactSheet before delivery.
- Designed Hybrid Research Sub-Agent: FAISS vector DB over PDFs from AWS S3, LLM relevancy gating with live Tavily search + BeautifulSoup fallback, ReAct tool loop, and **LangSmith** evaluation on 4 LLM-as-judge metrics (correctness, concision, groundedness, retrieval relevance).

Company Database AI Manager — Natural Language MCP Database Agent GitHub

- Built a natural-language interface to a corporate SQLite database, letting users insert, update, and query records in plain English instead of writing SQL, served through a Streamlit web UI.
- Architected a true Server/Client system using the **Model Context Protocol (MCP)**: a FastMCP server exposes database tools (`setup_database`, `add_record`, `search_database`) over stdio, decoupling database logic from AI logic.
- Implemented a **LangGraph** agent powered by Groq's `llama-3.3-70b-versatile` / `openai/gpt-oss-120b` that autonomously discovers available MCP tools, reasons over user intent, and triggers the correct Python functions without direct imports.
- Modeled relational schemas (Departments, Roles, Employees, Projects) with **SQLAlchemy** and enforced secure key handling via `.env` and `.gitignore` to keep credentials and the database out of version control.

MLOps Heart Attack Prediction Pipeline GitHub

- Automated reproducible ML pipeline with **DVC** (preprocess → train → evaluate); RandomForestClassifier achieving 98.1% test accuracy and 0.994 ROC-AUC with hyperparameters in `params.yaml`.
- Deployed to AWS EC2 via **GitHub Actions** CI/CD (test → lint → rsync → Gunicorn restart); served via Flask REST API with zero manual steps post-commit.

Skills

Languages:	Python
AI/ML Frameworks:	PyTorch, TensorFlow/Keras, Scikit-learn, Hugging Face Transformers
Computer Vision:	NVIDIA DeepStream, InsightFace, YOLOv8, OpenCV, FAISS, RetinaFace, ONNX
Agentic AI:	LangGraph, LangChain, ReAct Agents, RAG Pipelines, LLM Guardrails, LangSmith
Generative AI:	Groq API, Gemini API, Tavily Search, SentenceTransformers, FLAN-T5
MLOps & Infra:	Docker, FastAPI, Flask, DVC, GitHub Actions, AWS EC2, AWS S3

Education

Integrated M.Sc. Data Science Amrita Vishwa Vidyapeetham *Aug 2021 – Apr 2026*

Research: LSTM-based AI for cloud infrastructure resiliency using Sierpinski Graph topology

Certifications

- **Generative AI: AI Agents & RAG** – IBM via Coursera
- **TensorFlow Keras Bootcamp** – OpenCV University (Grade: 92)
- **Computer Vision Bootcamp** – OpenCV University (Grade: 94)